Lecture 24 09/11/26

Warm-up

Find the solutions of the following parametric linear system for all values of $k \in \mathbb{R}$:

$$\begin{cases} x + y = 2 \\ kx + y = k+1 \end{cases}$$

Solution

1) First of all, we recognize two equations and two unknowns. This is a 2×2 parametric linear system.

2) We built the coefficient matrix and the augmented matrix:

$$A = \begin{pmatrix} 1 & 1 \\ k & 1 \end{pmatrix} \qquad A|b = \left(\begin{array}{cc|c} 1 & 1 & 2 \\ k & 1 & k+1 \end{array}\right)$$

COEFFICIENT MATRIX

AUGMENTED MATRIX

3) Then we find the maximum values for the ranks:

- $1 \leq rk(A) \leq 2$

The rank $rk(A) \neq 0$ because $A \neq 0$, where 0 is the null matrix.

At the same time A is a 2×2 matrix, so $\max(rk(A)) = \min(\#\text{rows}, \#\text{columns}) = 2$.

- For the augmented matrix we know that

$$rk(A) \leq rk(A|b) \leq \min(\#\text{rows}, \#\text{columns}) = \min(2, 3) = 2$$

4) Finding the $rk(A|b)$ using Gauss-Jordan elimination. In this way we'll find the pivots of $A|b$.

$$A|b = \left(\begin{array}{cc|c} 1 & 1 & 2 \\ k & 1 & k+1 \end{array}\right)$$

$$\sim \quad 2^{nd}\text{ row} - k \times 1^{st}\text{ row} \quad \left(\begin{array}{cc|c} 1 & 1 & 2 \\ 0 & 1-k & -k+1 \end{array}\right)$$

- If $k = 1$: $A|b \sim \left(\begin{array}{cc|c} \textcircled{1} & 1 & 2 \\ 0 & 0 & 0 \end{array}\right)$

we get only one pivot. This means $r(A|b) = 1$.

- If $k \neq 1$: $A|b \sim \left(\begin{array}{cc|c} \textcircled{1} & 1 & 2 \\ 0 & \textcircled{1-k} & -k+1 \end{array}\right)$ 2 pivots

$rk(A|b) = 2$.

5) We study the particular case $k=1$, $rk(A|b) = 1$. This means that

$$1 \leq rk(A) \leq rk(A|b) \Rightarrow rk(A) = 1.$$

By the Rouché-Capelli theorem the linear system is compatible.

We have $rk(A) = 1$ with 2 unknowns. We have to introduce a free parameter $t \in \mathbb{R}$.

$$\left(\begin{array}{cc|c} 1 & 1 & 2 \\ 0 & 0 & 0 \end{array}\right) \Rightarrow \begin{cases} x + y = 2 \\ y = t \in \mathbb{R} \end{cases}$$

$$A' = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad \det A' = 1$$

$$A'_x = \begin{pmatrix} 2 & 1 \\ t & 1 \end{pmatrix}, \quad \det A'_x = 2 - t$$

$$A'_y = \begin{pmatrix} 1 & 2 \\ 0 & t \end{pmatrix}, \quad \det A'_y = t$$

$$x = \frac{\det A'_x}{\det A'} = \frac{2-t}{1} = 2 - t$$

$$y = \frac{\det A'_y}{\det A'} = \frac{t}{1} = t$$

$$(x, y) = (2 - t, t), \quad t \in \mathbb{R}, \quad k = 1.$$

When $k \neq 1$ we get

$$A|b \sim \left(\begin{array}{cc|c} 1 & 1 & 2 \\ 0 & 1-k & 1-k \end{array}\right) \quad rk(A|b) = 2$$

and also the $rk(A) = 2 \Rightarrow \begin{pmatrix} 1 & 1 \\ 0 & 1-k \end{pmatrix}$.

By Rouché-Capelli theorem we get a compatible system.

2 equations, 2 unknowns $\Rightarrow$ 1 solution

$$\begin{cases} x + y = 2 \\ (1-k)y = 1-k \end{cases} \Rightarrow \begin{cases} x + y = 2 \\ y = 1 \end{cases}$$ for any $k \neq 1$.

This means $x = 2 - y = 2 - 1 = 1$

For $k \neq 1$ $(x, y) = (1, 1)$

Remarkable limits

Quick recall: as $x \to 0$:

$$\frac{\sin(ax)}{x} \to a \quad , \quad \frac{1-\cos(ax)}{x^2} \to \frac{a^2}{2}$$

$$\frac{e^{ax}-1}{x} \to a \quad , \quad \frac{\ln(1+ax)}{x} \to a$$

$$\frac{(1+ax)^b-1}{x} \to ab \quad , \quad (1+ax)^{1/x} \to e^a$$

EX 1

$$\lim_{x\to 0} \frac{\sin(5x)}{3x} = \frac{1}{3}\lim_{x\to 0} \frac{\sin(5x)}{x} = \frac{5}{3}\lim_{x\to 0} \frac{\sin(5x)}{5x}$$

$y = 5x$, $y \xrightarrow[x\to 0]{} 0$

$$= \frac{5}{3}\lim_{y\to 0} \frac{\sin(y)}{y} = \frac{5}{3}$$

$$\lim_{y\to 0} \frac{\sin(y)}{y} = 1.$$

EX 2

$$\lim_{x\to 0} \frac{1-\cos(4x)}{2x^2} = \frac{1}{2}\lim_{x\to 0} \frac{1-\cos(4x)}{x^2}$$

$$= \frac{1}{2}\,\frac{4^2}{2} = \frac{16}{4} = 4$$

EX 3

$$\lim_{x\to 0} \frac{e^{3x}-1}{\sin(2x)} = \lim_{x\to 0} \frac{e^{3x}-1}{x}\cdot\frac{x}{\sin(2x)}$$

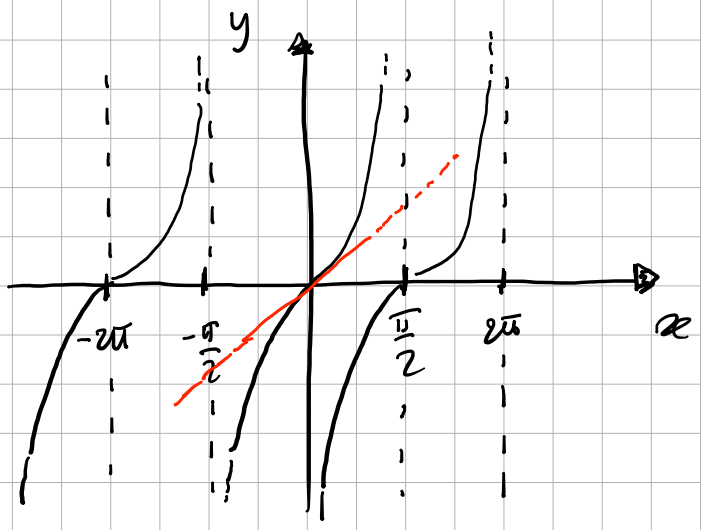
$$= \underbrace{\lim_{x\to 0} \frac{e^{3x}-1}{x}}_{3} \cdot \underbrace{\lim_{x\to 0} \frac{1}{\frac{\sin(2x)}{x}}}_{1/2} = \frac{3}{2}$$

EX 4 $\lim_{x\to 0} \frac{\ln(1+4x)}{e^{2x}-1} = \underbrace{\lim_{x\to 0} \frac{\ln(1+4x)}{x}}_{4} \cdot \underbrace{\lim_{x\to 0} \frac{x}{e^{2x}-1}}_{1/2} = 2.$

New limits

① The tangent function is a 2π-periodic function, and it is defined such as

$$\tan(x) = \frac{\sin(x)}{\cos(x)}.$$



$$\lim_{x\to 0} \frac{\tan(x)}{x} = 1.$$ Why? Because $\cos(0) = 1$ and $\lim_{x\to 0} \frac{\sin(x)}{x} = 1.$

So $$\lim_{x\to 0} \frac{\tan(x)}{x} = \lim_{x\to 0} \frac{\sin(x)}{\cos(x)} \cdot \frac{1}{x} = \lim_{x\to 0} \frac{\sin(x)}{x} \cdot \lim_{x\to 0} \frac{1}{\cos(x)} = 1 \cdot 1 = 1.$$

Generalizing, for every $a \in \mathbb{R}$, we get

$$\lim_{x\to 0} \frac{\tan(ax)}{x} = a$$

and for $b \neq 0$

$$\lim_{x\to 0} \frac{\tan(ax)}{bx} = \frac{a}{b}.$$

EX: $\lim_{x\to 0} \frac{\tan(3x)}{\sin(5x)} = \lim_{x\to 0} \frac{\tan(3x)}{x} \cdot \frac{x}{\sin(5x)} =$

$= \lim_{x\to 0} \frac{\tan(3x)}{x} \cdot \lim_{x\to 0} \frac{x}{\sin(5x)}$

$= 3 \cdot \frac{1}{5} = \frac{3}{5}.$

② let $b > 0$.

$$b^{ax} = e^{ax\ln(b)}.$$

this is always true, because, using the log properties we get:

$$e^{ax\ln(b)} = e^{\ln(b^{ax})} = \exp\left[\ln(\bullet)\right] = \bullet$$

↳ $e^{a\ln(b)\cdot x}$

$= b^{ax}.$

Therefore $\lim_{x\to 0} \frac{b^{ax}-1}{x} = \lim_{x\to 0} \frac{e^{a\ln(b)\cdot x}-1}{x} =$

Using $\lim_{x\to 0} \frac{e^{ax}-1}{x} = a$ ⟶ $= a\ln(b)$

Special case:

if $a = 1$: $\lim_{x\to 0} \frac{b^x - 1}{x} = \ln(b).$

EX: $\lim_{x\to 0} \frac{2^{3x}-1}{e^{4x}-1} = \underbrace{\lim_{x\to 0} \frac{2^{3x}-1}{x}}_{3\ln(2)} \cdot \underbrace{\lim_{x\to 0} \frac{x}{e^{4x}-1}}_{1/4} = \frac{3\ln(2)}{4}$

Asymptotic equivalence near zero

When $\frac{f(x)}{g(x)} \to 1$ in the limit $x \to 0$, we say that $f(x)$ is asymptotically equivalent to $g(x)$.

In this case we write

$$f(x) \sim g(x) \quad (\text{as } x \to 0)$$

From the remarkable limits we know that

$$\sin(ax) \sim ax$$

$$\tan(ax) \sim ax$$

$$1 - \cos(ax) \sim \frac{a^2 x^2}{2}$$

$$e^{ax} - 1 \sim ax$$

$$\ln(1+ax) \sim ax$$

$$(1+ax)^b - 1 \sim abx$$

EX: $$\lim_{x \to 0} \frac{(e^{2x} - 1)\ln(1+3x)}{\sin^2(4x)}.$$

Near 0 we know $(e^{2x} - 1) \sim 2x$

$\ln(1+3x) \sim 3x$, $\sin(4x) \sim 4x$

This means

$$\frac{(e^{2x}-1)(\ln(1+3x))}{\sin^2(4x)} \sim \frac{2x \cdot 3x}{(4x)^2}$$

$$= \frac{6x^2}{16x^2} = \frac{3}{8}$$

So the $\lim_{x\to 0} \frac{(e^{2x}-1)\ln(1+3x)}{\sin^2(4x)} = \frac{3}{8}$.

De l'Hôpital's rule / Taylor

If we consider

$\lim_{x\to 0} \frac{e^x - 1 - x}{x^2}$, using the asymptotic behaviors we get $e^x - 1 \sim x$:

$$\lim_{x\to 0} \frac{(e^x-1)-x}{x^2} = \lim_{x\to 0} \frac{x-x}{x^2} = \frac{0}{0}.$$

This trick doesn't solve the problem. The first order x, for the function $e^x - 1$, doesn't delete the indeterminate form.

We have to introduce two new tools: De l'Hôpital & Taylor.

De l'Hôpital's rule to solve $\frac{\infty}{\infty}$ and $\frac{0}{0}$

Theorem $\frac{0}{0}$: Suppose $f(x)$ and $g(x)$ are differentiable near a (not necessarily at a), and

$$f(x) \to 0, \quad g(x) \to 0 \quad \text{near } a,$$

and $g'(x) \neq 0$ near a.

If $\lim_{x \to a} \frac{f'(x)}{g'(x)} = L$ exists (finite or infinite), (the same)

then $\lim_{x \to a} \frac{f(x)}{g(x)} = L$.

$\to \frac{f'(x)}{g'(x)} \neq \left(\frac{f(x)}{g(x)}\right)'$ we have to apply the derivative separately to $f(x)$ and $g(x)$.

Theorem $\frac{\infty}{\infty}$: Using the same hypothesis, when

$$|f(x)| \to \infty, \quad |g(x)| \to \infty \quad \text{near } a$$

we get

$$\lim_{x \to a} \frac{f(x)}{g(x)} = \lim_{x \to a} \frac{f'(x)}{g'(x)}$$

You can use it for finite points a, or for one-sided limits $a^{\pm}$, or for infinite limits $x \to \pm\infty$.

When can we use de l'Hôpital? When the hypothesis hold and for the form $\frac{0}{0}$ and $\frac{\infty}{\infty}$.

For the other cases: $0\cdot\infty$, $\infty-\infty$, 1^{∞}, 0^{∞} but you can transform them in $0/0$ or ∞/∞ we cannot apply it.

EX: $\lim_{x\to 1} \frac{\ln(x)}{x-1} = \frac{0}{0}$ / $\frac{d}{dx}\ln(x)$ and $\frac{d}{dx}(x-1)$ exist near 1.

So $\lim_{x\to 1} \frac{\ln(x)}{x-1} = \lim_{x\to 1} \frac{\frac{d\ln(x)}{dx}}{\frac{d(x-1)}{dx}} = \lim_{x\to 1} \frac{\frac{1}{x}}{1} = 1$.

EX $\lim_{x\to 0} \frac{e^{2x}-1}{\sin(3x)} = \frac{0}{0}$

$\lim_{x\to 0} \frac{e^{2x}-1}{\sin(3x)} = \lim_{x\to 0} \frac{2e^{2x}}{3\cos(3x)} = \frac{2}{3}$

EX $\lim_{x\to 0} \frac{e^{x}-1-x}{x^2} = \frac{0}{0}$

$\lim_{x\to 0} \frac{e^{x}-1-x}{x^2} = \lim_{x\to 0} \frac{e^{x}-1}{2x} = \frac{0}{0}$

We apply again de l'Hôpital

$\lim_{x\to 0} \frac{e^{x}-1-x}{x^2} = \lim_{x\to 0} \frac{e^{x}-1}{2x} = \lim_{x\to 0} \frac{e^{x}}{2} = \frac{1}{2}$.

EX ∞/∞

$$\lim_{x\to\infty} \frac{\ln x}{x} = \frac{\infty}{\infty}$$

$$\lim_{x\to\infty} \frac{\ln x}{x} = \lim_{x\to\infty} \frac{1/x}{1} = \lim_{x\to\infty} \frac{1}{x} = 0.$$

EX $\lim_{x\to+\infty} \frac{x^3}{e^x} = \frac{\infty}{\infty}$

1st application

$$\lim_{x\to+\infty} \frac{x^3}{e^x} = \lim_{x\to+\infty} \frac{3x^2}{e^x}$$

apply again

$$= \lim_{x\to\infty} \frac{6x}{e^x} =$$

apply again

$$= \lim_{x\to\infty} \frac{6}{e^x} = 0.$$

EX $0\cdot\infty$ form

$$\lim_{x\to 0^+} x\ln(x) = 0\cdot(-\infty)$$

$$\lim_{x\to 0^+} x\ln(x) = \lim_{x\to 0^+} \frac{\ln(x)}{1/x} = \frac{-\infty}{\infty}$$ ✓

Now we can apply de l'Hôpital

$$\lim_{x\to 0^+} x\ln(x) = \lim_{x\to 0^+} \frac{\ln(x)}{1/x} = \lim_{x\to 0^+} \frac{1/x}{-1/x^2} =$$

$$= \lim_{x\to 0^+} \frac{1}{x}\cdot(-x^2) = -\lim_{x\to 0^+} x = 0.$$

EX $\infty - \infty$ form

$$\lim_{x \to +\infty} \left(\sqrt{x^2+x} - x \right)$$

Here we can use directly the algebra

Rationalizing we get

$$\lim_{x \to +\infty} \left(\sqrt{x^2+x} - x \right) = \lim_{x \to +\infty} \frac{\sqrt{x^2+x} - x}{\sqrt{x^2+x} + x} \left(\sqrt{x^2+x} + x \right)$$

$$= \lim_{x \to +\infty} \frac{(x^2+x) - x^2}{\sqrt{x^2+x} + x} = \lim_{x \to +\infty} \frac{x}{\sqrt{x^2+x} + x}$$

$$= \lim_{x \to +\infty} \frac{1}{\frac{\sqrt{x^2+x} + x}{x}} = \lim_{x \to +\infty} \frac{1}{\sqrt{1+\frac{1}{x}} + 1} = \frac{1}{2}$$

INDETERMINATE POWERS

$$1^{\infty} \quad / \quad 0^0 \quad / \quad \infty^0$$

1) compute $\lim (f(x))$
2) apply the limit
3) get the exponential of the limit

EX $\lim_{x\to+\infty} \left(1+\frac{a}{x}\right)^x$

$f(x) = \left(1+\frac{a}{x}\right)^x \longrightarrow \ln(f(x)) = x \ln\left(1+\frac{a}{x}\right)$

$\lim_{x\to+\infty} \ln(f(x)) = \lim_{x\to\infty} x \ln\left(1+\frac{a}{x}\right) = \infty \cdot 0$

$= \lim_{x\to\infty} \frac{\ln\left(1+\frac{a}{x}\right)}{1/x} = \frac{0}{0}$

de l'Hôpital

$= \lim_{x\to\infty} \frac{-\frac{a}{x^2}/\left(1+\frac{a}{x}\right)}{-1/x^2}$

$= \lim_{x\to\infty} \frac{-\frac{a}{x^2}\cdot(-x^2)}{1+\frac{a}{x}} = \lim_{x\to\infty} \frac{a}{1+\frac{a}{x}} = a$

Then $\lim_{x\to\infty} f(x) = e^{\lim_{x\to\infty} \ln f(x)} = e^a$

Taylor

We know that the first derivative represents the slope of the tangent line at a point of our function.

This means that near to this point we can approximate the function by the tangent line



$$f(x) \approx f(x_0) + f'(x_0)(x - x_0)$$

To get a better approximation, the Taylor expansion or Taylor polynomial works with derivatives of higher orders:

$$f(x) \approx P_n(x)$$ ← this is a polynomial with maximum degree n.

- Example: if $n=1$, we are approximating the $f(x)$ by the tangent line.

But if $n=2$, we are approximating $f(x)$ by a tangent parable, etc.

Taylor Polynomial of order n of f centered at x_0 is

$$P_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!} (x-x_0)^k$$

$f^{(k)}(x_0)$: derivative of order k computed in x_0

$k!$: factorial of k: $k! = k(k-1)(k-2)\cdots$

$$P_n(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{f''(x_0)}{2}(x-x_0)^2 + \frac{f'''(x_0)}{6}(x-x_0)^3 + \cdots$$

If $x_0 = 0$, it is called MACLAURIN POLYNOMIAL

$$f(x) = P_n(x) + R_n(x)$$

$R_n(x)$: this is the remainder

this measures the error of the approximation

$$R_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!}(x-x_0)^{n+1}, \quad \xi \in (x, x_0)$$

Maclaurin expansion of $f(x) = e^x$ (mean to $x_0 = 0$)

$$e^x = f(x_0) + f'(x_0)(x-x_0) + \frac{f''(x_0)}{2}(x-x_0)^2 + \dots$$

$$= f(0) + f'(0)x + \frac{f''(0)}{2}x^2 + \dots$$

$$f(0) = e^0 = 1$$

$$f'(0) = (e^x)'\big|_{x=0} = e^x\big|_{x=0} = e^0 = 1$$

$$f''(0) = (e^x)''\big|_{x=0} = e^x\big|_{x=0} = e^0 = 1$$

$$e^x = 1 + (x) + \frac{x^2}{2} + \dots$$

if we stop the approximation at the 1st order

$$e^x \approx 1 + x$$

at the second order : $e^x \approx 1 + x + \frac{x^2}{2}$

" third order : $e^x \approx 1 + x + \frac{x^2}{2} + \frac{x^3}{6} \dots$